Applied Data Mining

Final project

# Credit Risk Analysis Using Machine and Deep Learning Models

## Discussion and Insights

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Nowadays, with rapid development in Big Data technology, data availability, and computing power, banks or lending institutions are reviewing their business models for better decision-making. The paper discusses several algorithms to answer the question of whether to provide a loan to an enterprise, along with the strategies to determine variables, parameters, and criteria to evaluate models. Binary classifiers are built based on three machine learning models and four deep learning models on real data to predict loan default probability. The elastic net approach (logistic regression with regularisation, M1) is considered as a benchmark, as classical linear models are widely used in the banking industry. Two tree-based approaches, random forest model (M2) and gradient boosting machine (M3), as well as a neural network approach with four different complexities (D1, D2, D3, D4), are selected and compared.

The data set used in the paper contains 117,019 subjects (60% used for the fitting, 20% used for the cross-validation, and the rest 20% used for testing purpose) characterised by 181 variables. To analyse the variable importance and to test the stability of binary classifiers, the authors investigated the performance of these algorithms using only the top 10 important variables selected by each algorithm. The performances of models were evaluated using AUC (built from the ROC curve) and RMSE criteria. The paper concludes that the two tree-based models outperform the four deep learning models.

We would like to propose three points of interest which may potentially enhance the modelling evaluation process. Firstly, the data set used in the paper is only limited to static financial statement data. We suggest to include more macroeconomic factors and lagged terms as well. Secondly, the top 10 important variables selection process is a bit ambiguous in the context of model comparison. In our opinion, it would be better if models can be evaluated with variables selected by the stepwise method based on some specified criterion. Lastly, other criteria can also be used to evaluate and compare the models. For instance, the accuracy of prediction is a more intuitive and effective indicator.

## Data: The first learning basis

Pertaining to the data used, the information is mainly on the company's financial statements, balance sheets, income statement, and cash flow statement. However, macroeconomic variables have significant explanatory power to predict the probability of default. It has been shown that the model with macro conditions considered can better capture the absolute level of probability of default than the model with only firm-specific information[1].

Such variables include, but are not limited to:

- Unemployment rate
- Inflation
- GDP growth
- Industrial production growth
- Change in consumer sentiment
- Bank interest rate

[1] Kenneth Carling, Tor Jacobson, Jesper Linde, Kasper Roszbach (2007). Corporate credit risk modeling and the macroeconomy. Journal of Banking Finance 31 (2007) 845–868. doi:10.1016/j.jbankfin.2006.06.012

- Stock market performance

The rationale behind including macroeconomic variables is the fact that one shall expect high interest rate or slow economic growth to cumulatively lead to a gradual increase in credit risk.

In addition, lagged terms for variables can be included in the model building process. Basically, we can add the factors to the model with a lag function: $X_t = \frac{\sum_{k=1}^{K} \delta^{k-1} X_{t-k}}{\sum_{k=1}^{K} \delta^{k-1}}$ where $\{X_t : t = 1, , T\}$ is the raw monthly data series for a given variable. $X_t$ represents the value used in the model for the X series in month t, K is the length of the lag window, and $\delta$ is the decay factor (if $\delta$=1, there is no decay) [2].
Meanwhile, it should be noted that adding lagged values without constraints would lead to far too many coefficients to fit. Therefore, we impose a basic lag structure on the data, such that each variable is a weighted average over a fixed window, with exponentially declining weights. Also, we should be careful when including lagged terms, because autocorrelation in the error term can lead to high bias in the coefficients estimates. Hence, autocorrelaion should be taken into account.

## Variables selection

The second aspect is about variables. In fact, the paper uses the top 10 variables selected by distinct models (definitely, not the same top 10 variables for each of the models). However, they may not be the most appropriate ones. Furthermore, there is lack of evidence on the reason why these 10 variables are chosen in the paper. Actually, we find that AUC of M2 model with top 10 variables from M2 is smaller than that with top 10 variables from M3, D2, and D4, and RMSE is larger than that with variables from M3 and D2. The paper states that the profile of the selected top 10 variables from the model M3 is essential in deciding whether to provide a loan or not. And this has been considered as one of the reasons why tree-based models over-perform than others.

Feature selection is an important procedure for high-dimensional data to fit models. Hence, we suggest to select the variables by the stepwise method, one of the frequently-used evaluation measures. To be more precise, either forward selection or backward elimination method can be used. In the forward selection method, vector x is progressively built up, by starting with the empty set and adding one or a few features at a time; the backward elimination technique works in the opposite direction: starting with full set (of the 181 variables) and features are progressively eliminated from x one at a time[3].

## Models evaluation

Evaluating the models' performance is of great importance in any statistical analysis. In terms of criteria to rank the models with respect to the companies' probability of default, the ROC curve and AUC criteria as well as RMSE criteria are used in the paper. Since our

[2]Stephen Figlewski, Halina Frydman, Weijian Liang, (2012). Modeling the effect of macroeconomic factors on corporate default and credit rating transitions. International Review of Economics and Finance 21 (2012) 87–105. doi:10.1016/j.iref.2011.05.004

[3]Tautvydas Cibas, Françoise Fogelman Soulié, Patrick Gallinari Sarunas Raudys (1996). Variable selection with neural networks. Neurocomputing, Volume 12, Issues 2–3, 31 July 1996, Pages 223-248. https://doi.org/10.1016/0925-2312(95)00121-2

objective is to know whether a bank can provide a loan to the enterprise, the prediction value can be easily evaluated via accuracy computed on the data set.

Generally speaking, if the data set is imbalanced, the use of accuracy is of no importance. However, as our data has been balanced to be 46% zero and 53% one by SMOTE algorithm, we can use $Accuracy(y, \hat{y}) = \sum_{i=1}^{m} sign(\hat{y}_i, y_i)$ as one of our evaluation metrics, where $sign(\hat{y}, y)$=1 when $\hat{y} = y$ and $sign(\hat{y}, y)$=0 otherwise. In our case, banks face two types of errors when providing a loan to an enterprise: to refuse a loan to a company when its probability of default is much lower than the one specified by the model (type I error: false refuse), or approve a loan to a company when its probability of default is much higher than the one specified by the model (type II error: false loan). So our accuracy could be defined as :

$$\frac{true\ \ loan + true\ \ false}{true\ \ loan + false\ \ loan + true\ \ refuse + false\ \ refuse}$$

By adding one additional criterion – Accuracy, we can have an intuitive sense on the accurate rate of our model. Thus the comparison of models is more comprehensive.

## Further insights

Further research directions which would be worth exploring, in the field of credit risk assessment in general, are:

- Use of pre-origination and post-termination data in order to gain better insights into the borrower's behaviour
- Collaboration between banks and other companies in order to access current employment and income information directly from the applicant's employer and make better mortgage loan decisions
- Inclusion of a sociological approach, along with the statistical analysis. Risk is not an "objective" fact in the business environment, but no steps have been taken, yet, to integrate sociological disciplinary assumptions and theories into this area. It is needed as banks are similar to individuals. They have their own perception of risk, and that perception changes to adapt to the society's requirements